ALEC XU

+1 248-550-1354 ⋄ alecx@umich.edu ⋄ https://alecxu00.github.io

RESEARCH INTERESTS

- Theoretical foundations of supervised deep learning and generative AI models.
- Efficient parameterizations and algorithms for transformers (LLMs, VLMs)

EDUCATION

University of Michigan, Ann Arbor
PhD Electrical and Computer Engineering — Expected Aug. 2027
Advisor: Dr. Qing Qu
GPA: 4.00/4.00

MSE Electrical and Computer Engineering — Dec. 2022
Advisor: Dr. Jeffrey Fessler
GPA: 4.00/4.00

BSE Electrical Engineering, Summa Cum Laude — Dec. 2021
GPA: 3.97/4.00

INDUSTRY EXPERIENCE

Apple — Sunnyvale, CA
Machine Learning Intern — May 2026 - Sept. 2026

- Developing real-time zero-shot instance segmentation methods.

MIT Lincoln Laboratory — Lexington, MA
Co-Op Research Student — Jan. 2023 - July 2023

- Developed novel edge detection reconstruction self-supervised learning (SSL) task for 3D axon segmentation. U-Nets pre-trained on proposed task achieved up to 5% improvement on downstream segmentation metrics (Dice score) over previous SSL tasks.

Summer Research Intern — May 2022 - Aug. 2022

- Developed and implemented novel two-stage deep learning pipeline for multi-channel signal detection and modulation recognition using YOLO object detector and DeepSig's ResNet convolutional network.

Summer Research Intern — May 2021 - Aug. 2021

JHU Applied Physics Laboratory — Laurel, MD
Summer Intern — May 2020 - Aug. 2020

H3D, Inc. — Ann Arbor, MI
Summer Intern — May 2019 - Aug. 2019

SELECT PUBLICATIONS & PREPRINTS

† Denotes equal contribution

[8] S. Kwon, **A. S. Xu**, C. Yaras, D. Song, L. Balzano, Q. Qu. "The Effect of Training Task Diversity for In-Context Learning through the Lens of Low-Dimensional Subspaces," under review at ACM JDS.

[7] **A. S. Xu**, C. Yaras, M. Asato, Q. Qu, L. Balzano. "Emergent Low-Rank Training Dynamics in MLPs with Smooth Activations," arXiv:2602.06208.

[6] S. Kwon†, **A. S. Xu**†, C. Yaras, L. Balzano, Q. Qu. "Out-of-Distribution Generalization of In-Context Learning: A Low-Dimensional Subspace Perspective," AISTATS 2026.

Short versions in AISTATS 2026 CauScale Workshop (**Best Paper**), NeurIPS 2025 WCTD Workshop.

[5] **A. S. Xu**, C. Yaras, P. Wang, Q. Qu. "Linearly Separable Features in Shallow Nonlinear Networks: Width Scales Polynomially with Intrinsic Data Dimension." AISTATS 2026.

[4] C. Yaras, **A. S. Xu**, P. Abillima, C. Lee, L. Balzano. "MonarchAttention: Zero-Shot Conversion to Fast, Hardware-Aware Structured Attention," NeurIPS 2025 (**Spotlight**).

[3] **A. S. Xu**, N. I. Shamsi, L. A. Gjesteby, L. J. Brattain. "Self-Supervised Edge Detection Reconstruction for Topology-Oriented 3D Axon Segmentation and Centerline Detection," WACV 2024.

[2] N. I. Shamsi, **A. S. Xu**, L. A. Gjesteby, L. J. Brattain. "Improved Topological Preservation in 3D Axon Segmentation and Centerline Detection using Geometric Assessment-driven Topological Smoothing (GATS)," WACV 2024.

[1] **A. S. Xu**, L. Balzano, J. A. Fessler, "HeMPPCAT: Mixtures of Probabilistic Principal Component Analysers for Data with Heteroscedastic Noise," ICASSP 2023.

AWARDS

Best Paper | AISTATS 2026 CauScale Workshop

Best Poster Runner-Up | University of Michigan ECE Industry Day

ECE Department Fellowship | University of Michigan

EECS Scholar | University of Michigan

James B. Angell Scholar | University of Michigan

William J. Branstrom Freshman Prize | University of Michigan

Dean's List/University Honors | University of Michigan

TEACHING

Graduate Student Instructor | University of Michigan
ECE 559: Optimization Methods for Signal Processing and Machine Learning 2026
ECE 551: Matrix Methods for Signal Processing & Machine Learning 2022, 2024

Undergraduate Instructional Aide | University of Michigan
EECS 455: Wireless Communication Systems 2021
EECS 398 / 498: Software Defined Radio 2021